

**JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)**

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 192

READ THE INSTRUCTIONS CAREFULLY**GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **20** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

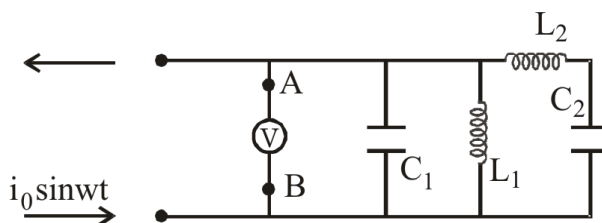
Space for Rough Work

PART-1 : PHYSICS

SECTION-II (i) : (Maximum Marks: 20)

- This section contains **FIVE (05)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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Full Marks : +4 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

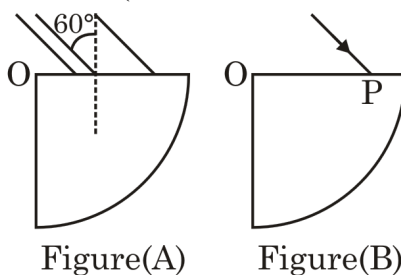
1. A source of current $i_0 \sin \omega t$, with i_0 as constant, is connected to the circuit shown in the diagram. The frequency ω is controllable.



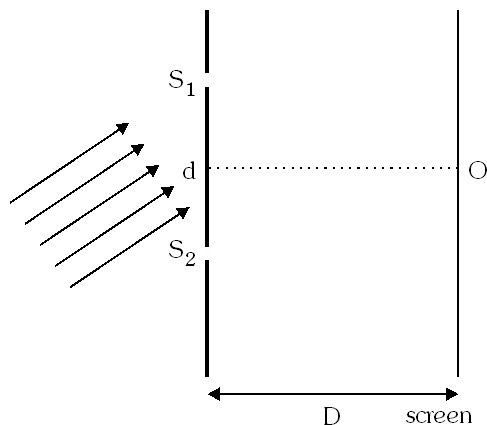
The inductors L_1 , L_2 and capacitor C_1 , C_2 are all ideal. An ideal voltmeter is connected between points A and B. For very small ω (but non zero), reading of voltmeter is $\frac{i_0 \omega L_1}{x}$. Find x.

2. The rubber shell of a balloon has mass $m = 585 \frac{\pi}{2}$ gm. We fill methane (CH_4) in it at room temperature. The molar mass of air can be assumed to be 29 gm/mol. The balloon always remains spherical in shape and the pressure difference between inside & outside due to its elastic nature can be neglected. At what diameter (in cm) will the balloon float in air? Thickness of balloon is negligible.
 $(p_{\text{atm}} = 10^5 \text{ Pa}, R = \frac{25}{3} \text{ J/mol K}, T_0 = 27^\circ\text{C}.)$

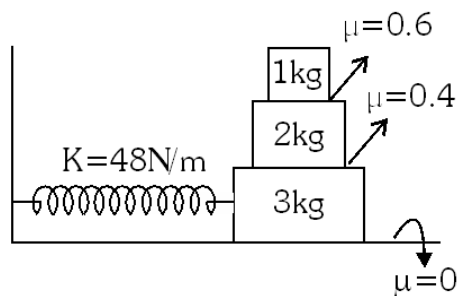
3. A parallel beam of light falls on a quarter cylinder of radius 9 cm as shown in figure (A). Refractive index of the material of the cylinder is $\sqrt{3}$. Maximum value of OP in meter shown in figure (B), so that rays don't suffer TIR at the curved surface is (consider refractive index of surrounding to be 1).



4. A monochromatic light of wavelength λ_0 and intensity I_0 is incident on YDSE experiment. The distance between slits S_1S_2 is 'd' ($d \ll D$). The angle made by parallel beam light with the normal to the plane of slits is $\sin^{-1} \left(\frac{3\lambda}{2d} \right)$. Now, if a transparent glass slab of thickness $\frac{3\lambda}{2(\mu - 1)}$ is kept in front of one of the slits, where ' μ ' is refractive index of glass slab, find the intensity (in W/m^2) at the geometrical centre 'O' of the screen. (Given, $d = 1\text{mm}$, $D = 2\text{m}$, $\mu = 4/3$, $\lambda = 4000 \text{ \AA}$ and $I_0 = 4.16 \text{ W/m}^2$)



5. What can be the maximum amplitude (in m) of the system so that there is no slipping between any of the blocks ? (Take $g = 10 \text{ ms}^{-2}$).



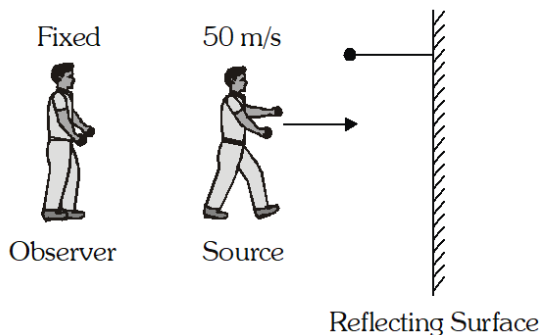
SECTION-II (ii) : (Maximum Marks: 20)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

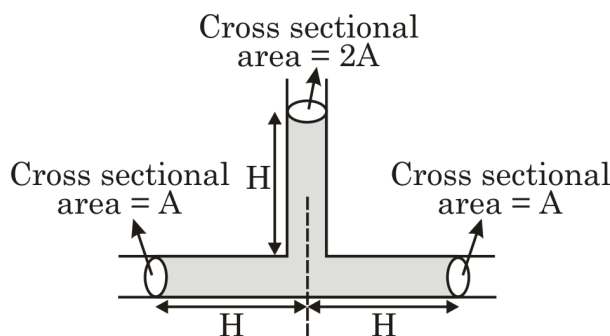
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

6. A source of sound frequency 1000 Hz moves to the right with a speed of 50 m/s relative to the ground. To its right is a reflecting surface moving to the left with a speed of 20 m/s. There is a stationary observer to the left of the source as shown in figure. Speed of sound in air is 330 m/s. Then wavelength of reflected wave (in mm) is :-

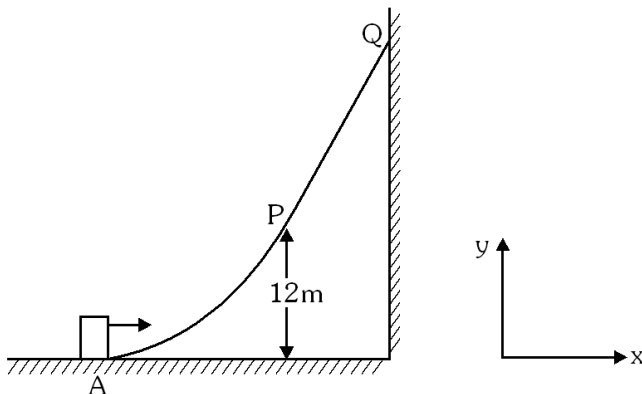


7. Figure shows inverted T shaped fixed pipe. Liquid is released from rest from the position shown. Liquid is non-viscous. Velocity of fluid particle in vertical part of tube when level of liquid in it remains $H/2$ will be $\sqrt{\frac{xgH}{8}}$. Find x .



8. A satellite weighing 3 tons flies in an approximately circular orbit near earth. The resistive force acting on it is 0.01 N. How much will its flight height (in km) decrease in a period of 10 day? (Round off to nearest integer) (Earth's radius = 6400 km, $g = 10 \text{ m/s}^2$)

9. APQ is a rough surface. The portion AP is parabolic given by equation $x^2 = 16y$. The point 'A' assumed to be origin. The portion PQ is a straight line of length $20\sqrt{3}$ m, the coefficient of friction between block and surface (APQ) is $\frac{1}{\sqrt{3}}$. A block of mass 3 kg is moved on surface APQ by applying a force on it. The direction of force is always tangential to the surface APQ. The line PQ is tangent on the curve AP at P. If the block moves very slowly on the surface then find the work done by applying force in joule till it reaches Q.



10. A 2000 W resistance heater wire has thermal conductivity 15 W/mK, radius of 4 mm and length 0.5 m. It is used to boil water. The outer surface of wire has temperature of 105°C . What is the temperature (in $^\circ\text{C}$) at the axis of the wire, if heat flows out only radially? Round off to nearest integer.

SECTION-III : (Maximum Marks: 24)

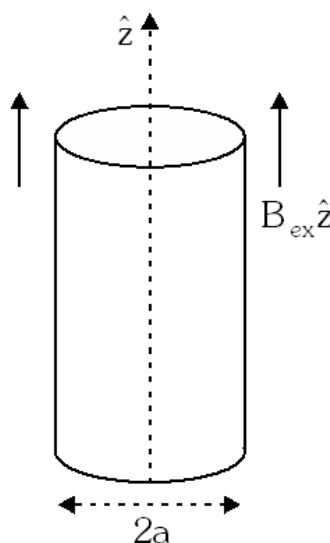
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Full Marks : +3 If ONLY the correct answer is chosen.

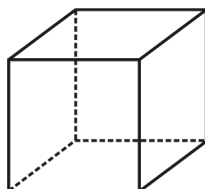
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

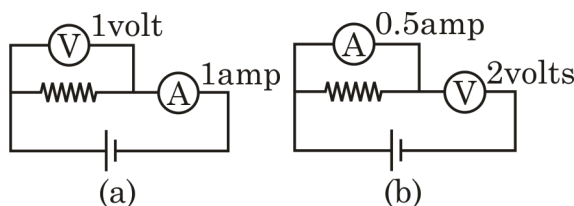
1. There is an insulating uniformly charged cylindrical shell of radius a lies with its axis along the z -axis. The shell's moment of inertia per unit length about the z -axis and the surface charge density are I and σ respectively. The cylinder is placed in an external uniform magnetic field $B_{ex}\hat{z}$, and is initially at rest. Starting at $t = 0$ the external magnetic field is slowly reduced to zero. If the final angular velocity ω of the cylinder is $\omega = \frac{\sigma\pi a^x B}{I} \left(1 + \frac{\mu_0 \sigma^y a^z \pi}{I}\right)^d$, find $x + y + z + 2d$.



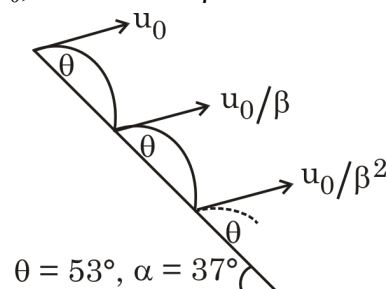
2. We have a cube whose 3 adjacent faces are charged uniformly with surface charge density of σ . The remaining 3 faces are charged with a surface charge density of $-\sigma$. If the electric field at the centre of the cube is $\frac{\sigma}{\sqrt{x}\epsilon_0}$, find x .



3. A sphere made of polymer is floating on the surface of water which is at a temperature of 20°C and 1% of the volume of the sphere is above the water. To what final temperature (in $^{\circ}\text{C}$) must the water be heated in order to submerge the polymer sphere completely ? Take coefficient of linear expansion of the polymer $2.30 \times 10^{-5}/^{\circ}\text{C}$, and coefficient of volume expansion of water $3.19 \times 10^{-4}/^{\circ}\text{C}$. If your answer is N, find value of $\frac{N}{20}$.
4. We want to measure the resistance of a resistor. A voltmeter and ammeter are connected as shown in figure a & b. Their readings are also indicated in figure. What is the voltage of the battery (in V)? The battery is ideal.

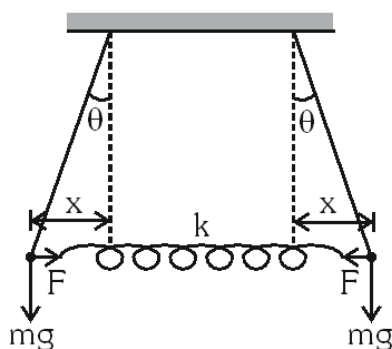


5. A ball is thrown from the inclined plane (large) at an angle θ from inclined surface with an initial speed u_0 . For the resulting projectile motion, after hitting the surface, ball rebounds at the same angle θ , but with reduced speed $\frac{u_0}{\beta}$. Its motion continues for a long time. If the magnitude of average velocity of the ball for entire duration of motion is u_0 , then value of β is :-

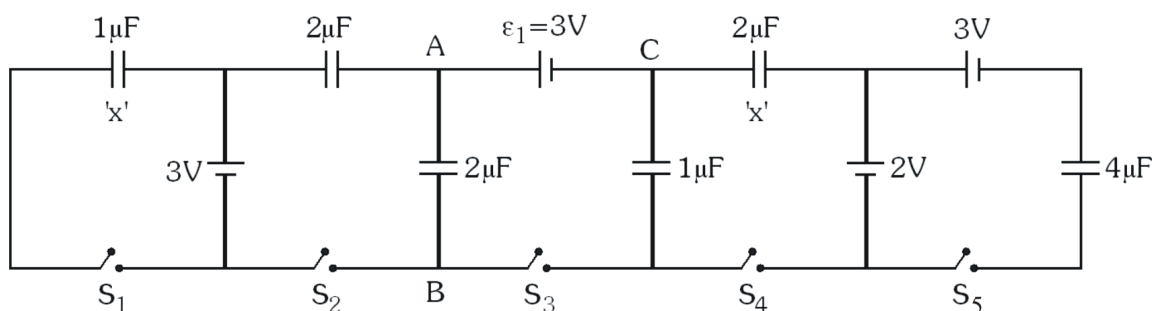


6. In the determination of Young's modulus by using Searle's method, a wire of length $L = 2\text{m}$ and diameter $d = 0.5\text{ mm}$ is used. For a load $M = 2.5\text{ kg}$, an extension $\ell = 0.25\text{ mm}$ in the length of the wire is observed. Quantities d and ℓ are measured using a screw gauge and a micrometer, respectively. They have the same pitch of 0.5 mm . The number of divisions on their circular scale is 100. The length of wire, acceleration due to gravity and mass M are known exactly. The percentage error in measurement of the Young's modulus is _____.

7. Two identical simple pendulums each of length ℓ are connected by a weightless spring as shown in figure. The force constant of the spring is k . In equilibrium, the pendulums are vertical and the spring is horizontal and underformed. If the time period of small oscillation of the linked pendulums, when they are deflected from their equilibrium positions through equal displacements in the same vertical plane in opposite direction and released is $2\pi \sqrt{\frac{\ell}{\left(g + \frac{nk\ell}{m}\right)}}$, find n .



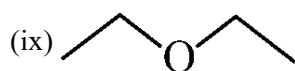
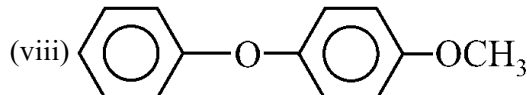
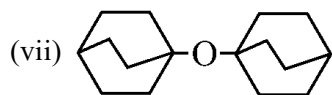
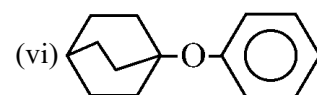
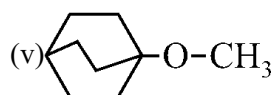
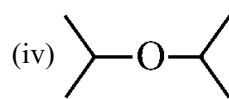
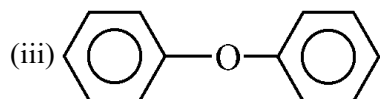
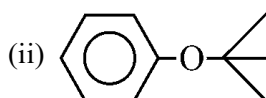
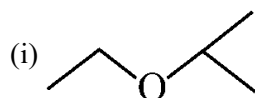
8. Six capacitors and four ideal batteries are connected in a circuit as shown in the figure. Initially all capacitors are uncharged and all switches were open. Now, all switches are closed simultaneously. Work done by $\varepsilon_1 = 3V$ battery is _____ (in μJ) :



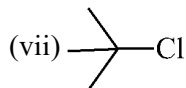
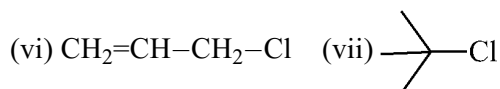
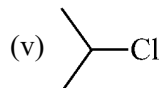
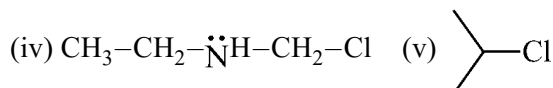
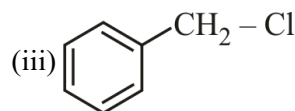
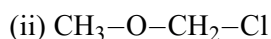
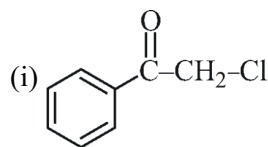
PART-2 : CHEMISTRY
SECTION-II (i) : (Maximum Marks: 20)

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- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. How many ether(s) will undergo cleavage on reaction with conc.HI ?



2. How many compounds are more reactive towards $\text{S}_\text{N}1$ as well as $\text{S}_\text{N}2$ than $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--Cl}$:



3. At 300 K temperature and 8 atm pressure the compressibility factor for a real gas is 0.8. The volume (in litre) of 10 mole of the gas at same temperature and pressure is
[Given : $R = 0.0821 \text{ L-atm mol}^{-1} \text{ K}^{-1}$]
4. The rate constant for the reaction $\text{N}_2\text{O}_4(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ is $92.12 \times 10^{-2} \text{ min}^{-1}$. The time (in seconds) required for the molar ratio of $\text{NO}_2(\text{g})$ to $\text{N}_2\text{O}_4(\text{g})$ in the reaction mixture to become 2 : 1 is
[$\log 2 = 0.3010$, $\ln 10 = 2.303$]
5. Sum of the spin only magnetic moment (in the units of B.M.) of the following complexes:
 $[\text{Cu}(\text{NH}_3)_4]^{2+}$; $[\text{Fe}(\text{CN})_6]^{3-}$; $[\text{NiCl}_4]^{2-}$; $[\text{CoF}_6]^{3-}$; $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$
[Given: $\sqrt{2} = 1.41$, $\sqrt{3} = 1.73$, $\sqrt{5} = 2.24$, $\sqrt{7} = 2.64$, $\sqrt{15} = 3.87$, $\sqrt{24} = 4.90$]

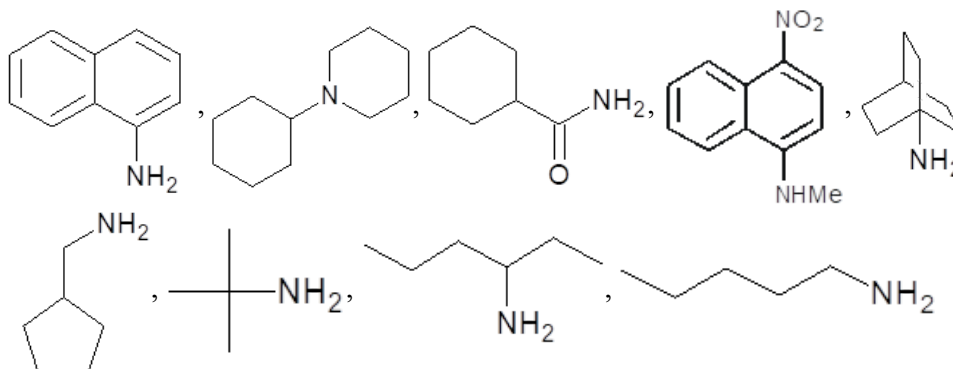
SECTION-II (ii) : (Maximum Marks: 20)

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Zero Marks : 0 In all other cases

6. How many are correctly matched for purification of organic compounds?
- Distillation is used for separation of chloroform & aniline.
 - Fractional distillation for crude oil in petroleum industry.
 - Distillation under reduced pressure is used for liquid having very high boiling point.
 - Distillation under reduced pressure is used for spent-eye in soap industry.
 - Steam distillation for aniline water mixture.
 - Paper chromatography is type of adsorption chromatography.
 - Column chromatography is type of paper chromatography.
 - Thin layer chromatography is type of adsorption chromatography.
7. How many of the following can be prepared by gabriel phthalimide synthesis in good yield.



8. An ionic compound is created with anions A & B present at corners of cube in 3 : 1 ratio. All face-centers & body centre contain cation Z. Atomic weight of A, B and Z are respectively 20, 40 & 15 amu. Edge length of unit cell is 1 Å, density = $d(\text{amu } \text{\AA}^{-3})$ and empirical formula of ionic compound is $A_aB_bZ_c$, here a, b and c are smallest integer value. Then value of (a + b + c + d) considering unit cell is electrically neutral.

9. 1 mole of an ideal monoatomic gas initially at 1 atm & 300 K undergoes a process by which pressure is doubled. The nature of the process is unspecified but for the process : $\Delta U = 900$ cal. Calculate final volume of gas (in litres) ?

[Given: $R = 0.08$ atm lit./mol/K = 2 cal / K-mol]

10. Calculate total number of coloured compound (white/colourless not include) out of the following

- | | | | |
|---|------------------------------------|--|---|
| (1) FeS | (2) FeS ₂ | (3) Zn ₂ [Fe(CN) ₆] | |
| (4) Fe ₃ [Fe(CN) ₆] ₂ | (5) Cu ₂ I ₂ | (6) ZnS | (7) Ag ₂ S ₂ O ₃ |
| (8) Ag ₂ S | (9) ZnO(Hot) | (10) CrO ₂ Cl ₂ | (11) Cu ₂ [Fe(CN) ₆] |
| (12) Hg ₂ CrO ₄ | (13) KO ₃ | (14) KBr | (15) H ₂ S |

SECTION-III : (Maximum Marks: 24)

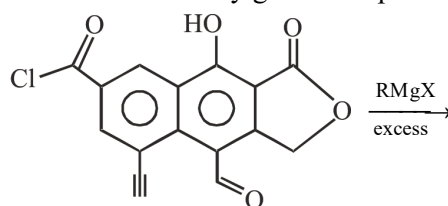
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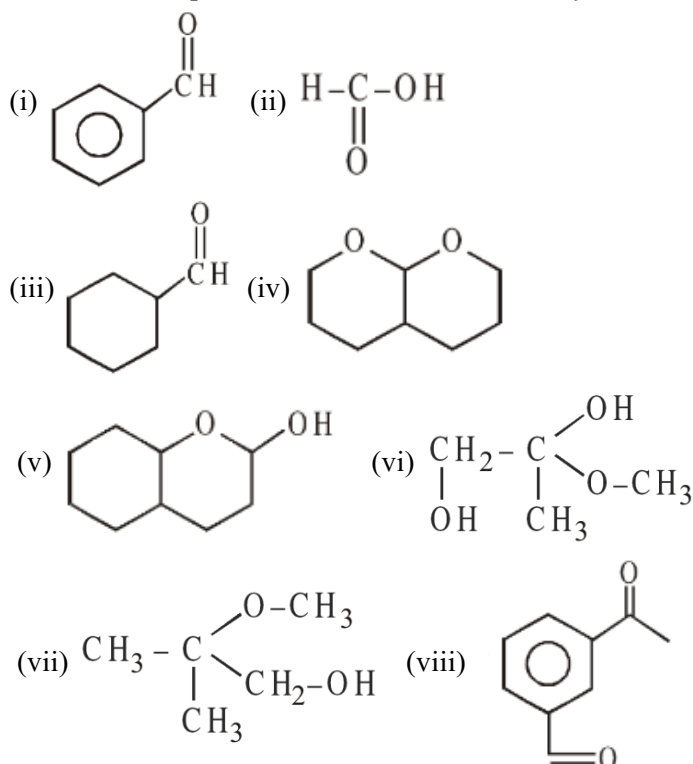
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. Number of Grignard reagent that can be consumed by given compound.



2. Number of compounds which can be oxidised by Tollen's reagent.



3. The mole fraction of solute in some solution is $1/n$. If 50% of the solute molecules dissociates into two parts and remaining 50% get dimerized now mole fractions of solvent become $4/5$. Find value of n .

4. An aqueous solution has 2 mole of NH_4Cl and this solution is filled between two electrodes which are 20 cm apart. Resistance of this solution is measured to be 100Ω . Calculate molar conductance ($\text{S cm}^2 \text{ mole}^{-1}$) for NH_4Cl .
5. In electrolytic refining of impure copper containing impurities of Ag, Hg, Al, Zn, Au, Fe, Pt, Sn, Mg. Then find out total number of metals present in anode mud in their elemental form.
6. How many of the following will give $\text{N}_2(\text{g})$ on heating.
 $(\text{NH}_4)_2\text{SO}_4$, NH_4NO_3 , $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$, NH_4Cl , NH_4NO_2 , NH_4MnO_4 .
7. Find the total stereoisomers for :
 (a) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-} = x$
 (b) $[\text{PtCl}_2(\text{en})_2]^{2+} = y$
 (c) $[\text{Cr}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+ = z$
 Hence write the value of $(x + y + z)$.
8. Anglesite ore + water $\rightarrow$ insoluble

$$\begin{array}{ccccc} \text{Anglesite} & + & \text{ammonium acetate} & \xrightarrow{\text{Boiling}} & \text{Soluble metal} & + & \text{other} \\ \text{ore} & & \text{solution (aq.)} & & \text{containing compound} & & \text{products} \\ & & & & (\text{X}) & & \end{array}$$
 Write the number of oxygen atoms in water soluble metal containing compound X.

PART-3 : MATHEMATICS
SECTION-II (i) : (Maximum Marks: 20)

- This section contains **FIVE (05)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

- If $f(x) = \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{\tan\left(\frac{x}{2^{r+1}}\right) + \tan^3\left(\frac{x}{2^{r+1}}\right)}{\left(1 - \tan^2 \frac{x}{2^{r+1}}\right)}$, then $\frac{\lim_{x \rightarrow 0} \frac{f(x)}{x}}{2}$ is :-
- Let the base of a pyramid is rectangular, three of its vertices of base are A(2, 2, -1), B(3, 1, 2) and C(1, 1, 1). Its vertex at the top is $P\left(4, \frac{-26}{3}, \frac{-10}{3}\right)$. Let the coordinates of foot of normal drawn from P on the base of pyramid is (ℓ , m, n) then $\frac{\ell + m + n}{10}$ is :-
- $\frac{\sum_{r=1}^{\infty} \tan^{-1}\left(\frac{6 \cdot 7^{(r-1)}}{1+7^{(2r-1)}}\right)}{\pi}$ is :-
- If $\sec A \tan B + \tan A \sec B = 91$, then $\frac{(\sec A \sec B + \tan A \tan B)^2}{4141}$ is
- For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - (3 + 4i)| = 5$, the minimum value of $|z_1 - z_2|$ is :-

SECTION-II (ii) : (Maximum Marks: 20)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

6. The remainder when 7^{103} is divided by 25 is :-
7. Let $f(x) = \max\{x^2, (1-x)^2, 2x(1-x)\}$, where $0 \leq x \leq 1$. If the area bounded by $y = f(x)$, x-axis, $x = 0$ & $x = 1$ is λ , then $27(\lambda)$ is :-
8. There are 10 tickets marked numbers 1,2,3,...,10. Three tickets marked numbers a, b, c ($a > b > c$) are drawn at random. If the probability that $\int_0^a (x^2 - 2bx + 3c)dx = 0$ is $\frac{1}{p}$, then the value of 'p' is
9. A circle of radius 1 is randomly placed in a 9×12 rectangle ABCD so that the circle lies completely within the rectangle. If $\frac{m}{n}$ (where m and n are relatively prime numbers) denotes the probability that the circle neither touches nor intersect the diagonal AC, then the value of $(m + n)$ is
10. Let S_n be the area of the figure enclosed by a curve $y = x^2(1-x)^n$ ($0 \leq x \leq 1$) and x-axis, if $\lim_{n \rightarrow \infty} \sum_{k=1}^n S_k = \frac{1}{\sqrt{A}}$, then 'A' is equal to

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. The sum of $\frac{3}{1.2} \left(\frac{1}{2}\right) + \frac{4}{2.3} \left(\frac{1}{2}\right)^2 + \frac{5}{3.4} \left(\frac{1}{2}\right)^3 + \dots \infty$ terms is
2. Let $(1+x)^{10} = a_0 + a_1x + a_2x^2 + \dots + a_{10}x^{10}$, then $\frac{(a_0 - a_2 + a_4 - a_6 + a_8 - a_{10})^2 + (a_1 - a_3 + a_5 - a_7 + a_9)^2}{2^{10}}$ is
3. If $f(x+y) = f(x) + f(y) - xy - 1 \quad \forall x, y \in \mathbb{R}$ and $f(1) = 1$, then number of solutions of $f(x) = x, x \in \mathbb{N}$ is
4. The shortest distance between z-axis and the line $x + y + 2z - 3 = 0 = 2x + 3y + 4z - 4$ is :-
5. Let $2y \sin x \frac{dy}{dx} = 2 \sin x \cos x - y^2 \cos x$ satisfying $y \left(\frac{\pi}{2}\right) = 1$, then $\frac{y^2}{\sin x}$ (wherever defined) is :-
6. $\lim_{x \rightarrow 0} \frac{\left(\frac{a^x + b^x + c^x}{3}\right)^{2/x}}{(abc)^{2/3}}$ is :-
(Where $a, b, c > 0$)
7. Let $A = \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}$ and $(A + I)^{50} - 50A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $a + b + c + d$ is
8. If z_1 & z_2 be two distinct complex numbers satisfying $|z_1^2 - z_2^2| = |\bar{z}_1^2 + \bar{z}_2^2 - 2\bar{z}_1\bar{z}_2|$ & if $(\arg z_1 - \arg z_2) = \frac{a\pi}{b}$, then least possible value of $|a+b|$ is, ($a, b \in \text{Integer}$)

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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E-20/20

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